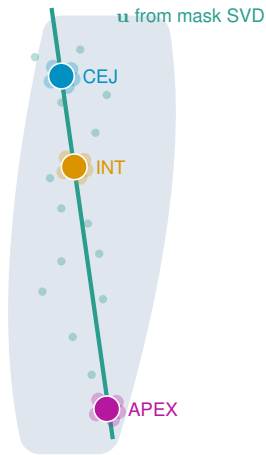


Reference Eq. 1 — the axis is fitted *to the three landmarks*

$\ell = \ell(\mathbf{c}, \mathbf{i}, \mathbf{a})$, so a perturbation ε moves the points *and* turns the line they are projected onto:

$$\frac{\partial S}{\partial \varepsilon} = \underbrace{\frac{\partial S}{\partial \mathbf{p}}}_{\text{displacement}} + \underbrace{\frac{\partial S}{\partial \ell} \frac{\partial \ell}{\partial \varepsilon}}_{\text{rotation, non-zero}}$$

Measured consequence. Injecting isotropic Gaussian noise of known σ into ground-truth landmarks and re-measuring against each formula's own clean value (v6 labels, 200 test images, 10 realisations per σ): the mask-PCA definition is the more robust at *every* non-zero noise level, and the margin widens monotonically — +0.014 at $\sigma=4$ px, +0.018 at 8px, +0.041 at 16px, +0.051 at 24px.



Mask-PCA axis — the axis is fixed *before* the landmarks are read

$\mathbf{u}$ is the principal component of thousands of mask pixels, so $\partial \mathbf{u} / \partial \varepsilon = 0$ and only the scalar coordinates move:

$$\frac{\partial S}{\partial \varepsilon} = \underbrace{\frac{\partial S}{\partial \mathbf{p}}}_{\text{displacement}} + \underbrace{0}_{\text{no rotation term}}$$